DATA ANALYSIS AND VISUALIZATION · MIDTERM PROJECT

Should we change the launch message next week?

Social-media sentiment analysis for a new F&B product launch, using 1.6 million tweets from Sentiment140

Group VibeCoding_Expert

Ngo Lam Tien (leader) · Nguyen Cong Thanh · Nguyen Le Nhat Tam · Ngo Binh Minh

Lecturer: Ho Thi Linh, Ph.D. · Ton Duc Thang University · 16 October 2026

THE DECISION

One business question, four analytical questions

Marketing needs a yes/no decision within days. We broke it down with the four types of analytics from Chapter 1, and fixed four hypotheses before looking at any result.

1 **Descriptive** **What happened?**

How big is the F&B conversation, what is it about, and how negative is it compared with all tweets?

2 **Diagnostic** **Why did it happen?**

Which aspects of the experience and which moments go together with negative tweets?

3 **Predictive** **What will happen?**

Can a model trained on past weeks label next week's tweets reliably?

4 **Prescriptive** **What should we do?**

Which message changes, how much impact, and when should the team be alerted?

Hypotheses: H1 service & waiting · H2 price & value · H3 morning caffeine differ from the rest of F&B · H4 launch-week lift differs from baseline. Practical threshold: ≥ 5 pp.

EXECUTIVE SUMMARY

Keep the core message, fix three pain points

−5.1 pp

F&B tweets are less negative than all tweets (negative lift)

p = 0.37

Launch week is no different from the baseline weeks (H4)

0.896

ROC-AUC of our model on the unseen next week

3

product issues to fix: availability, service, price

Recommendation

- No rewrite: the mixed reactions are the normal mix of the category, not a launch backlash.
- Targeted fixes where tweets are most negative: availability (+45.3 pp), price (+5.6 pp), service (+5.4 pp, evening +10.2 pp).
- Creative around trying and sharing; daily monitoring with an alert limit of +1.5 pp.

DATA

1.6 million tweets, one usable F&B stream

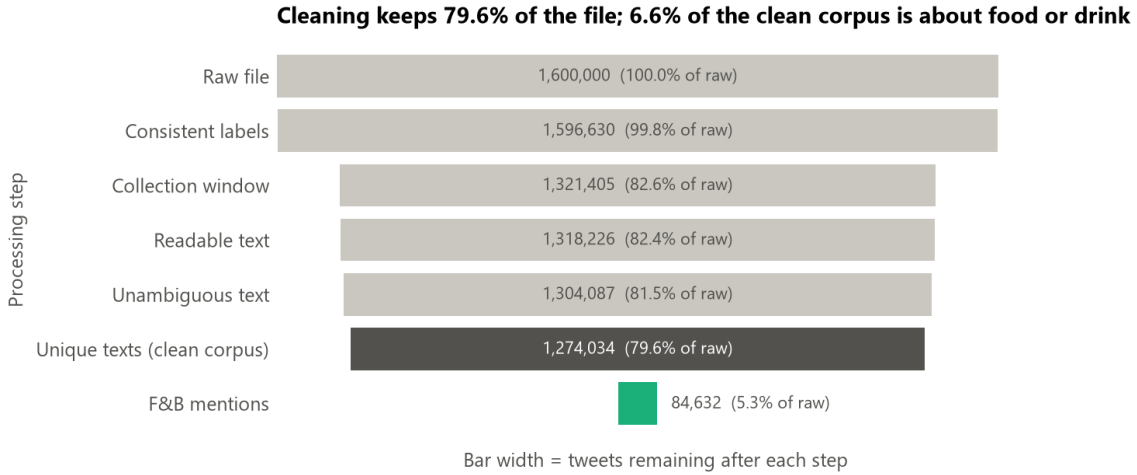
Source Sentiment140 (Go et al., 2009), Kaggle copy: 1,600,000 tweets, Apr-Jun 2009, 659,775 users

Labels From emoticons (0 = negative, 4 = positive); no neutral class; emoticons removed from the text

Coverage Only 48 of 81 days contain tweets; every timestamp is in Pacific time

Proxy Food & drink tweets stand in for the brand's category conversation: 6 categories, 41 brands

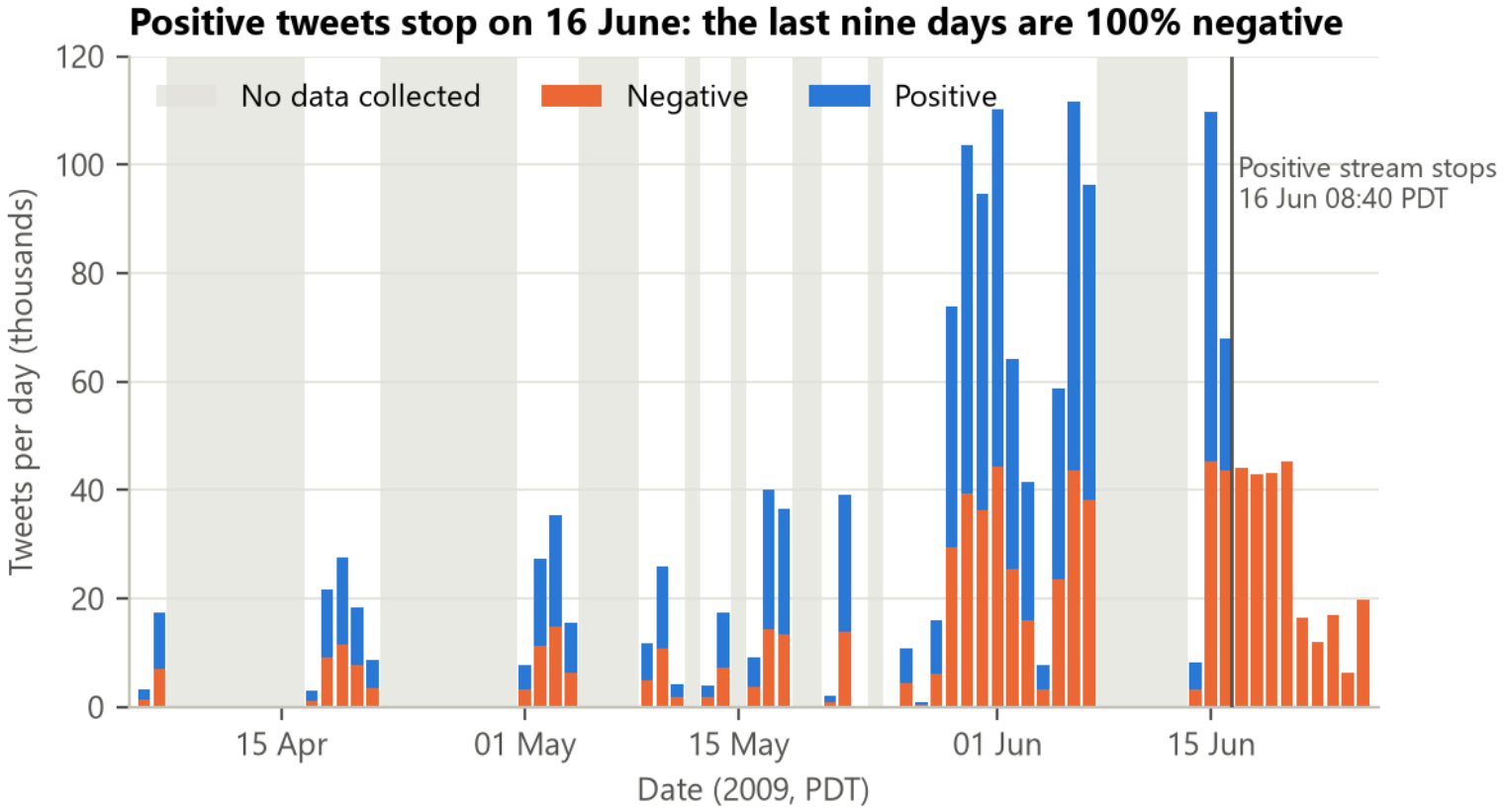
Check 210-tweet stratified review: dictionary precision 89.0% (95% CI 84.8-93.3%), AI-assisted



Cleaning keeps 1,274,034 tweets (79.6%); 84,632 of them are about food or drink.

DATA QUALITY

A trap in the file: positive tweets stop on 16 June



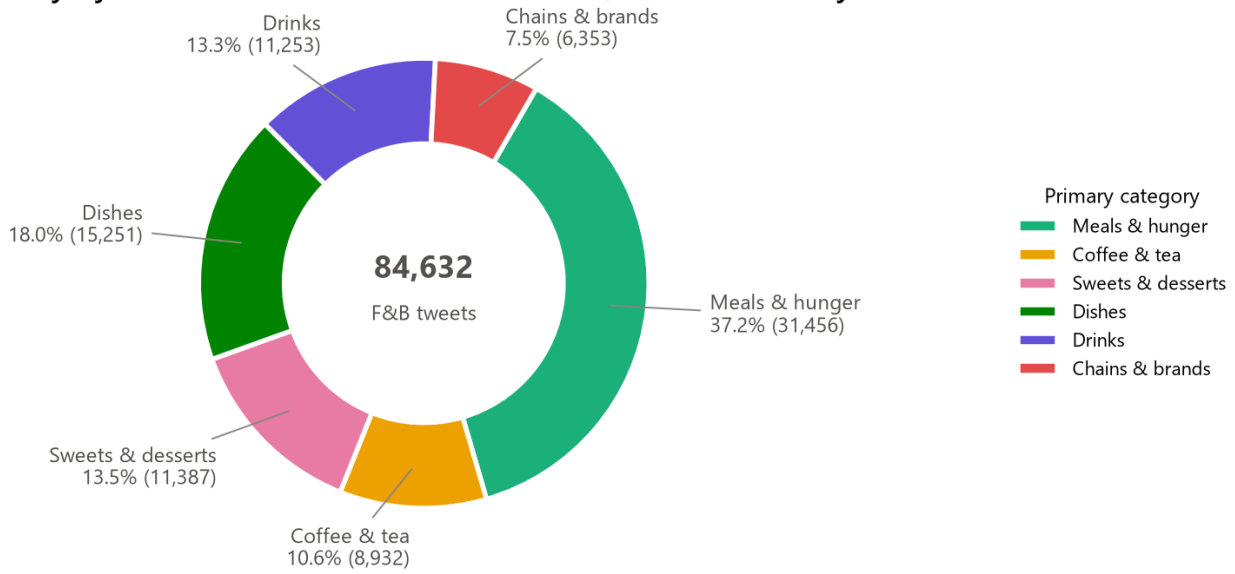
275,225

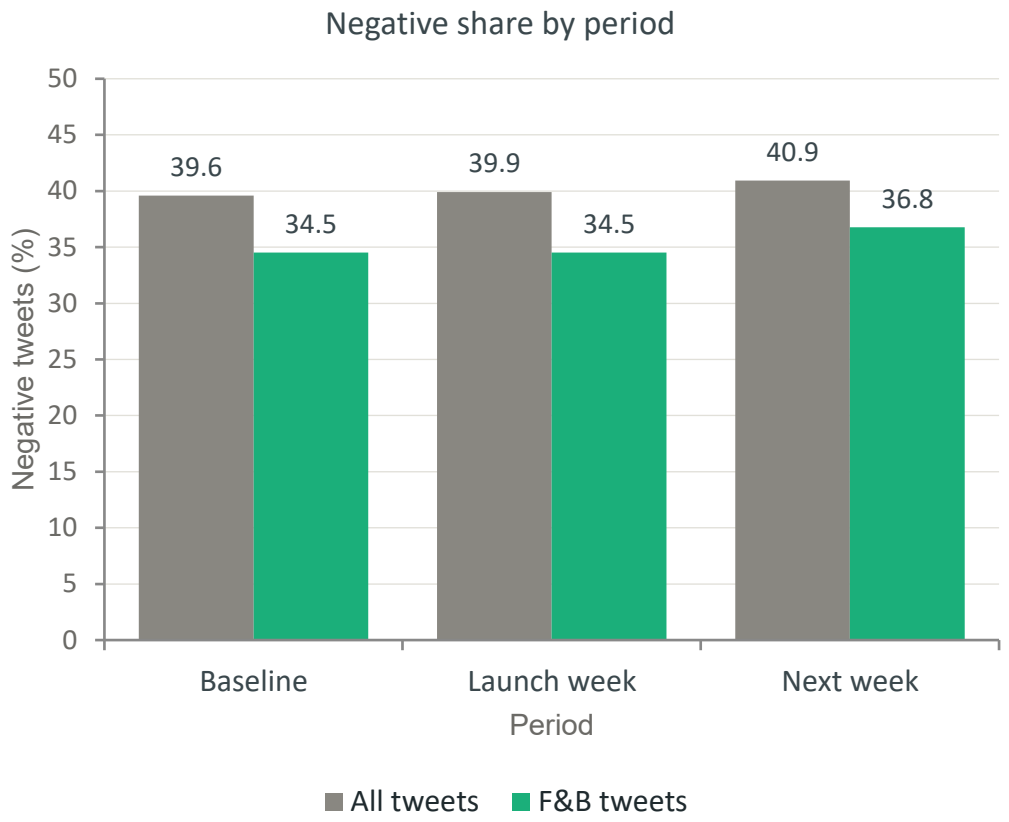
tweets after 16 June 08:40 PDT, and 100% of them are negative

- Removed from every analysis: keeping them would fake a sentiment collapse.
- Also removed: 1,685 contradictory ids and 30,053 repeated texts (bots).
- Positive and negative were collected separately, so we compare groups (lift, odds ratios), never absolute shares.

F&B talk is everyday, and less negative than Twitter

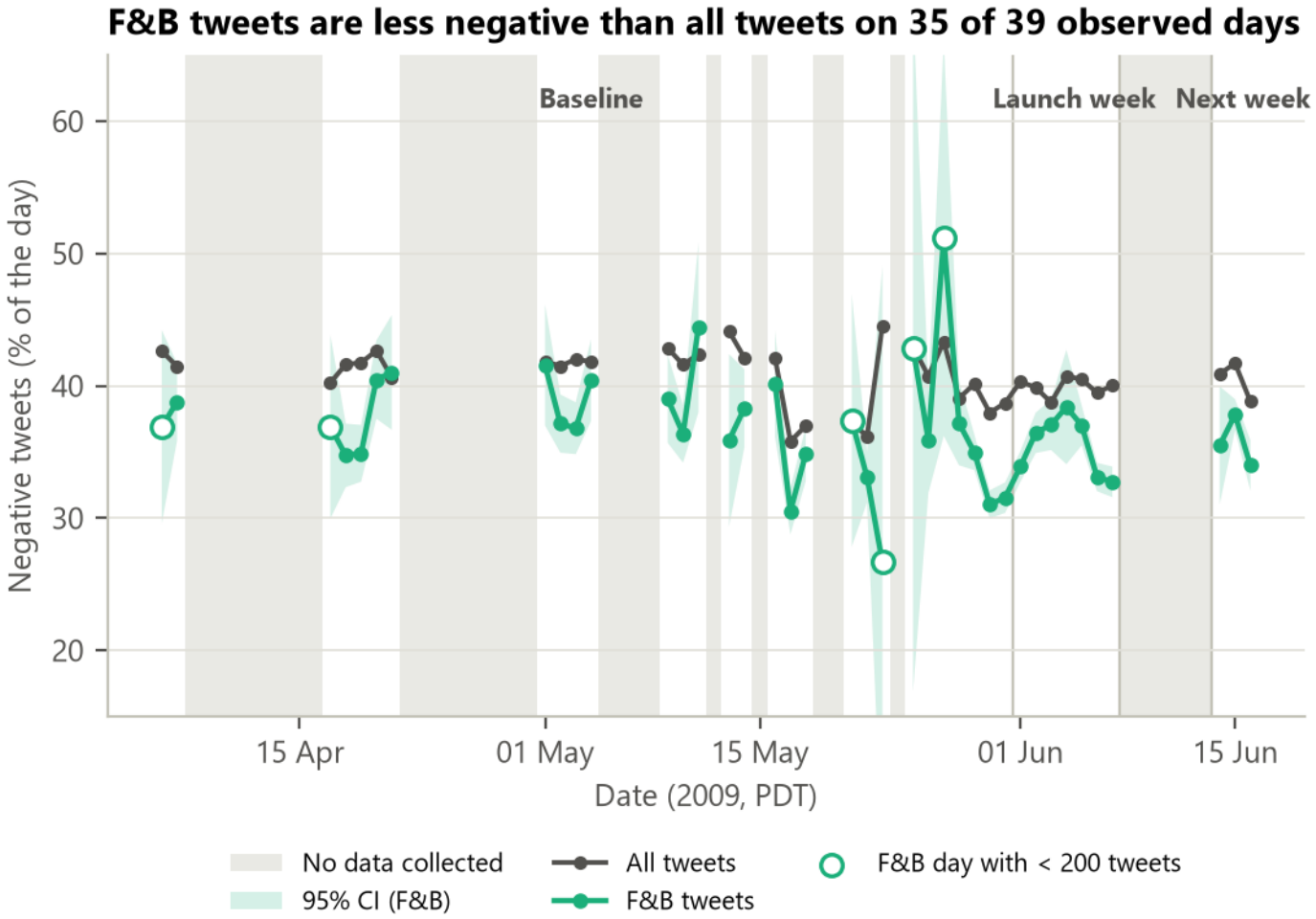
Everyday meal talk is 37% of the F&B conversation; named brands only 8%





Only 7.5% of F&B tweets name a brand. The F&B lift stays near −5 pp in every period (launch week −5.4 pp, baseline −5.1 pp).

Launch week looks like any other week

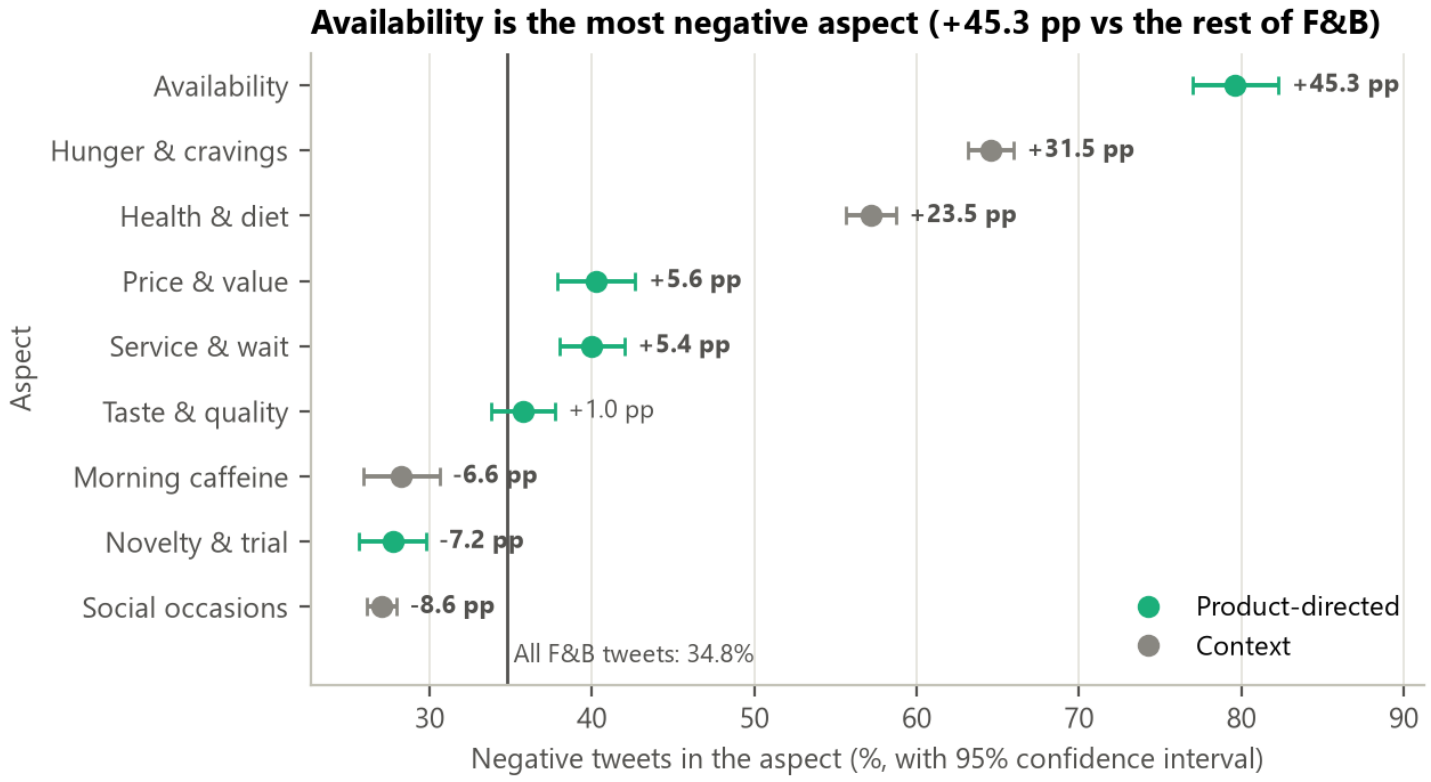


35 of 39

observed days with F&B less negative than all tweets

- The spikes are days with fewer than 200 F&B tweets: wide CIs, not events.
- Brands: 20 of 41 make up 80% of mentions; Starbucks alone 20%.
- Negative tweets are 1.3 words longer, but Cohen's d = 0.20 (small).

Availability, price and service drive negative talk

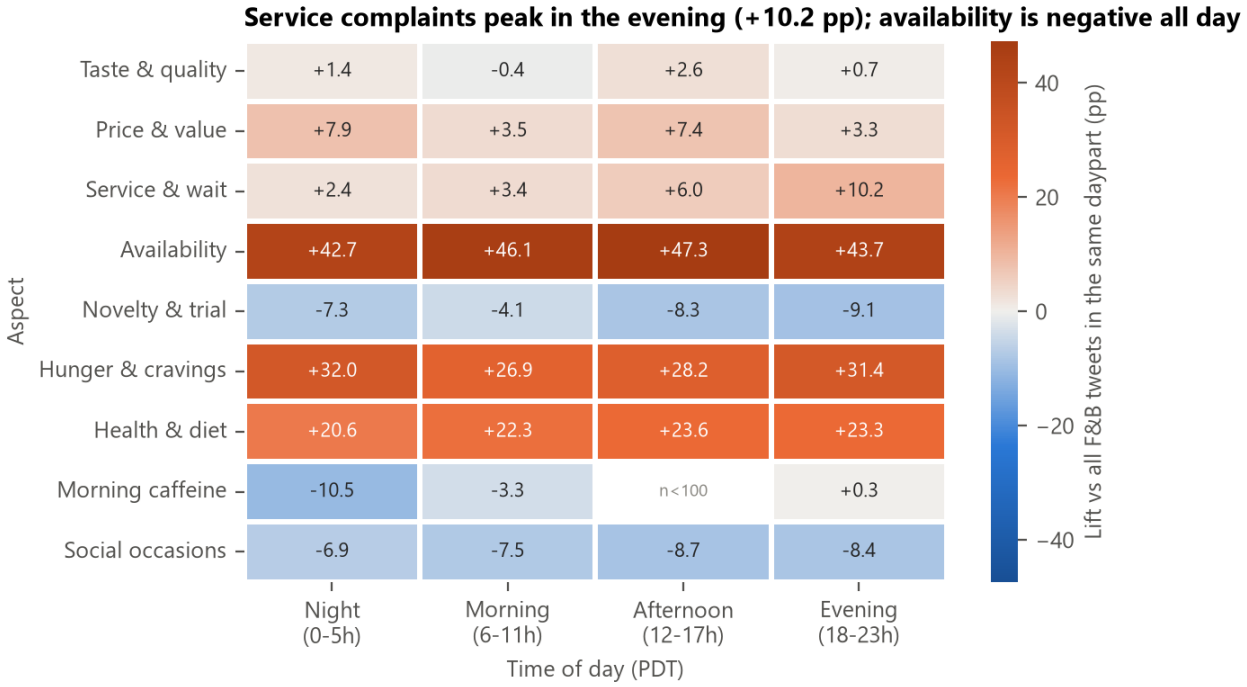


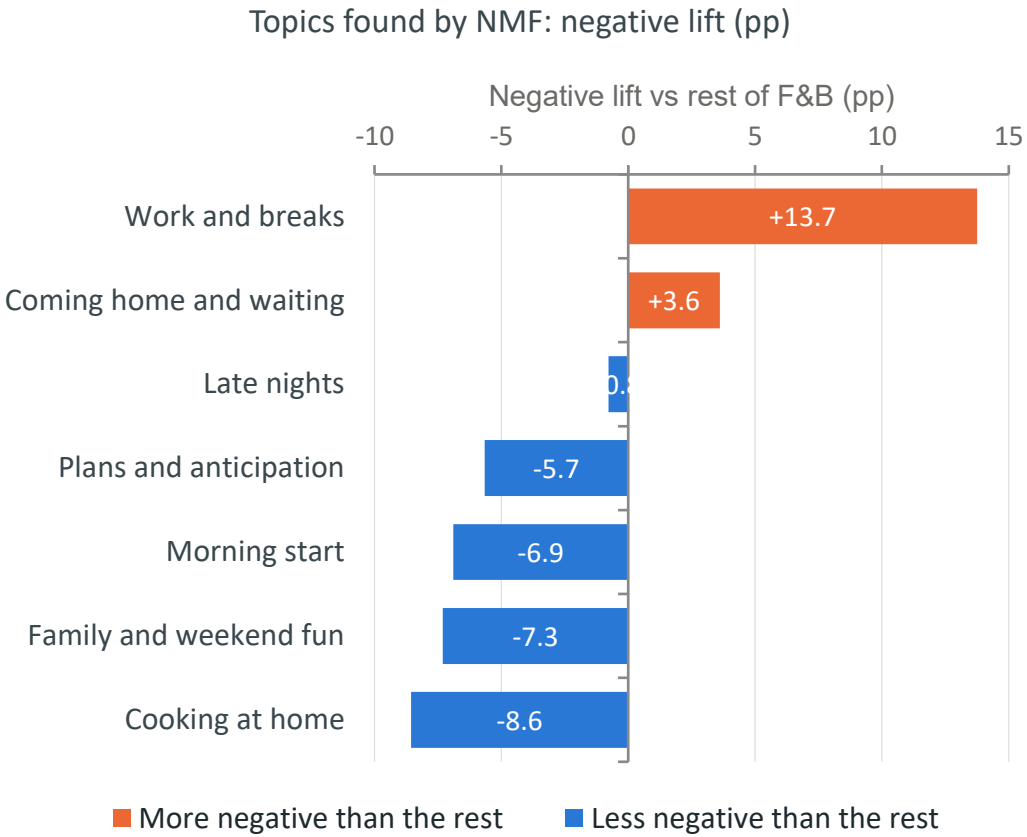
Hypotheses (Holm-corrected)

H1	**Service & wait**	+5.4 pp supported
H2	**Price & value**	+5.6 pp supported
H3	**Morning caffeine**	−6.6 pp supported
H4	**Launch vs baseline**	−0.3 pp, p = 0.37 not supported

Logistic regression (holding other factors and the day fixed): availability OR 7.2, novelty OR 0.66.

Complaints depend on the moment, not only the product





Service complaints peak in the evening (+10.2 pp); food around work and breaks is +13.7 pp, while home, family and morning moments are clearly positive.

PREDICTIVE ANALYTICS

Trained on the past, tested on next week

Train
6 Apr - 31 May · 653,345 tweets

Validation
1 - 7 Jun (launch week) · 469,953 tweets

Gap
8 - 13 Jun · no data

Test
14 - 16 Jun (next week) · 150,736, used once

Why split by time and not at random?

- Matches real use: the model is trained on the past and applied to next week.
- No leakage between days that share topics and conversations.
- Reveals drift; a random split hides it (Chapter 7: prediction refers to future values).

BASELINE

Multinomial Naive Bayes

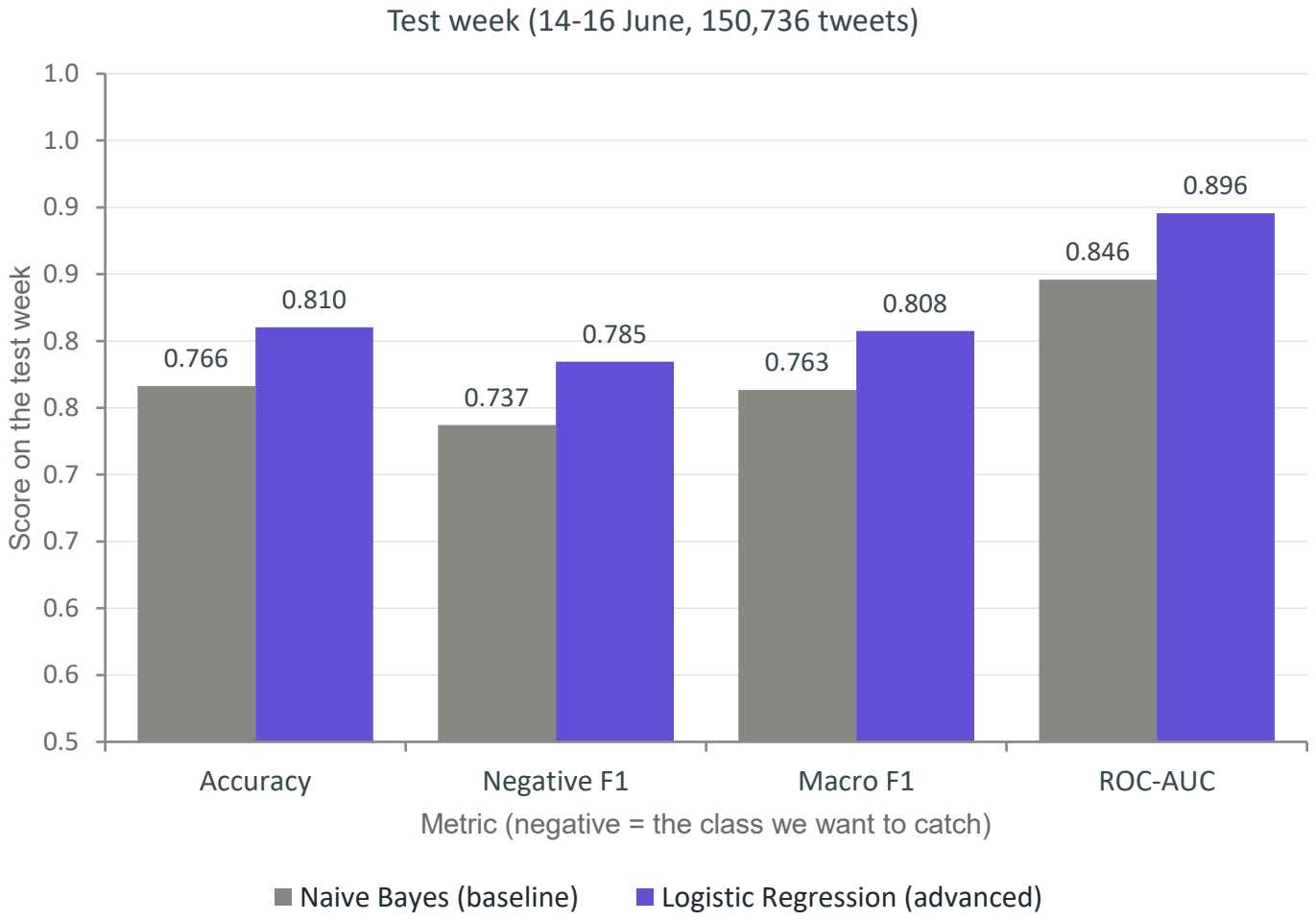
Word counts, 47,118 words, alpha tuned on validation (3.0). The classic baseline for this dataset.

ADVANCED

Logistic Regression

TF-IDF of words and word pairs (300,000 terms), C tuned (2.0), class weights, threshold 0.44 chosen on validation.

Logistic Regression wins on every metric



+4.4 pp

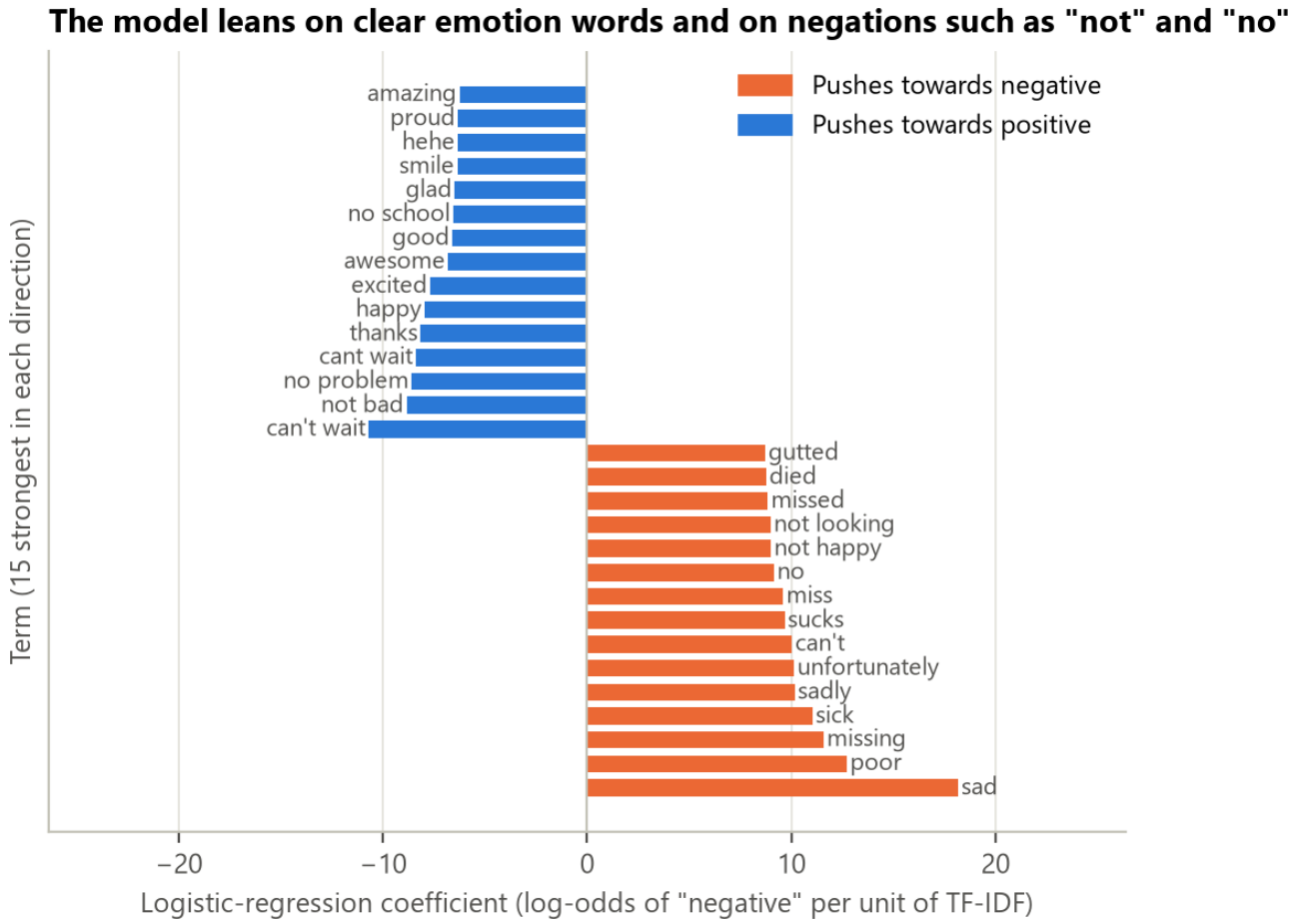
accuracy gain (McNemar, $p < 0.001$): about 440 fewer mislabelled tweets per 10,000

Robustness

- Random 80/20 split: AUC 0.903, a little optimistic.
- F&B tweets only: AUC 0.886.
- Model trained on F&B only: 0.862, so the general model is better.
- New users score the same as known users (0.895 vs 0.896).

MODEL INTERPRETATION

The model is readable, not a black box



Example 1: availability complaint

“i am so sad!! i ran out of orange juice boohoo!”

sad, boohoo, so sad, ran out push towards negative.
P(negative) = 1.00

Example 2: negation handled by word pairs

“that's not bad at all, as long as there's a smash cake…”

The pair “not bad” outweighs “not” and “bad”. P(negative) = 0.06

Decision rule: targeted adjustments, not a rewrite

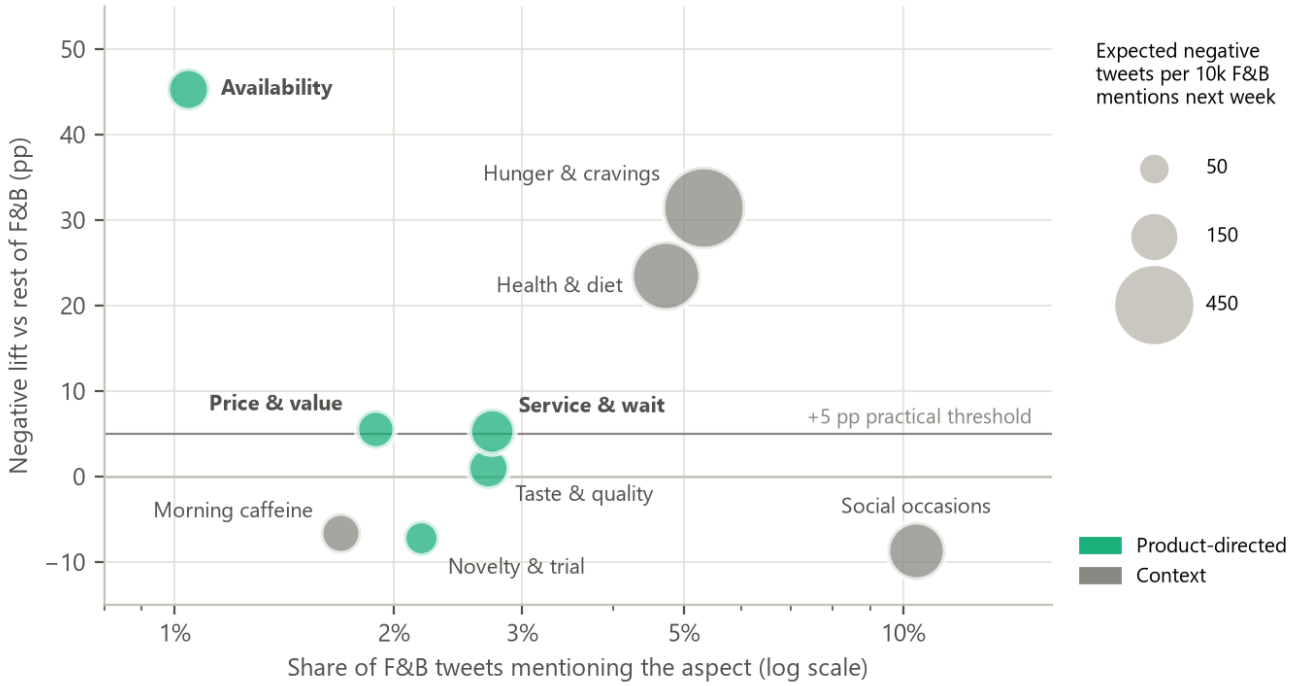
Rule written before the results

Rewrite the message
if launch week is significantly and practically more negative (H4)
Not met: H4 −0.3 pp, p = 0.37

Targeted adjustments
if a product aspect is ≥ 5 pp more negative, whole CI above 0
Met: availability, service, price

Keep and monitor
otherwise

Availability, price and service are the product issues worth acting on; hunger and health are context the brand cannot fix



PRESCRIPTIVE ANALYTICS

Six recommendations, each backed by numbers

R1 Keep the core message

Launch lift −5.4 pp ≈ baseline; H4 p = 0.37; next week estimated 38.2% (true 36.8%)

R2 Add an availability message

1.0% of tweets but 79.6% negative (+45.3 pp), OR 7.2; ~108 negative per 10k next week

R3 Fix service, evenings first

+5.4 pp (CI +3.3 to +7.4), evening +10.2 pp; ~129 negative per 10k next week

R4 Reframe price as value

+5.6 pp (CI +3.2 to +8.0), OR 1.21; trial offer, bundle or coupon

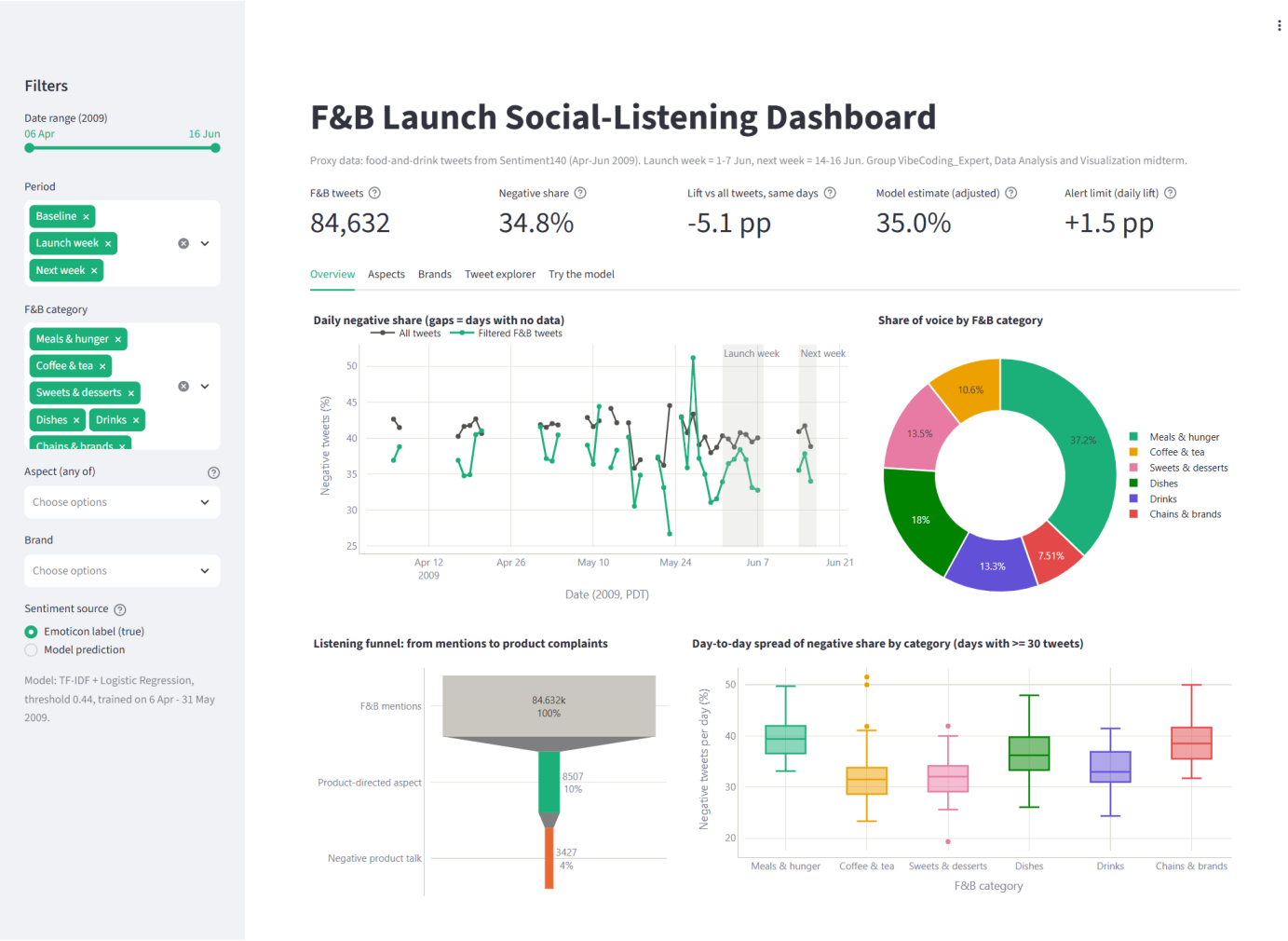
R5 Creative on trying and sharing

Novelty −7.2 pp, social −8.6 pp, morning −6.6 pp; avoid work-stress framing (+13.7 pp)

R6 Monitor daily

Classifier AUC 0.896, catches 84% of complaints; alert if daily lift > +1.5 pp

What-if, alert rule and a live dashboard



−0.7 pp

overall negativity if all three gaps were closed: act locally, set targets per aspect

+1.5 pp

alert limit (baseline mean + 2 SD): 1 false alarm in 23 days, 0 in launch or next week

Streamlit dashboard

Filters, KPI tiles, trend, aspects, brands, tweet explorer and a live “try the model” tab.

Keep the message, fix the moments that hurt

1. **Descriptive** F&B talk is everyday and 5.1 pp less negative than Twitter; launch week is normal.
2. **Diagnostic** Availability, price and service drive complaints; trying and sharing are positive.
3. **Predictive** Logistic Regression beats Naive Bayes on an unseen week: AUC 0.896 vs 0.846.
4. **Prescriptive** Targeted adjustments R2-R5 and daily monitoring with a +1.5 pp alert limit.

Limitations

- Proxy data from 2009, not one real launch
- Emoticon labels, no neutral class
- Collection gaps; Pacific-time stamps
- Filter precision 89% (AI-assisted check)
- Associations, not causes: confirm with an A/B test

Team Tien: problem, prescriptive, slides · Thanh: data, descriptive · Tam: diagnostic, conclusion · Minh: predictive, dashboard

Thank you. Questions?

APPENDIX A

Hypothesis tests H1-H4

	Comparison	Difference	95% CI (pp)	Holm p	Cohen's h	Verdict
H1	Service & wait vs rest of F&B	+5.4 pp	+3.3 to +7.4	< 0.001	+0.11	Supported
H2	Price & value vs rest of F&B	+5.6 pp	+3.2 to +8.0	< 0.001	+0.12	Supported
H3	Morning caffeine vs rest of F&B	−6.6 pp	-9.0 to -4.3	< 0.001	−0.14	Supported
H4	Launch-week lift vs baseline (diff-in-diff)	−0.3 pp	-1.0 to +0.4	0.37	n/a	Not supported

APPENDIX B

Robustness of Logistic Regression

Evaluation	Tweets	Accuracy	Negative F1	ROC-AUC
Main result: time split, all test tweets	150,736	0.810	0.785	0.896
Random 80/20 split (comparison only)	254,807	0.820	0.789	0.903
Time split, F&B test tweets only	9,527	0.811	0.758	0.886
Users already seen in training	88,298	0.811	0.778	0.896
Users never seen in training	62,438	0.809	0.793	0.895
Model trained on F&B tweets only	9,527	0.774	0.726	0.862

APPENDIX C

Cleaning log

Step	Rows before	Removed	Rows after	Reason
Consistent labels	1,600,000	3,370	1,596,630	Same id labelled positive and negative
Collection window	1,596,630	275,225	1,321,405	After 16 Jun 08:40 only negatives were collected
Readable text	1,321,405	3,179	1,318,226	No words left after removing links and users
Unambiguous text	1,318,226	14,139	1,304,087	Same normalised text with both labels
Unique texts	1,304,087	30,053	1,274,034	Repeated texts (bots, spam)